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Brain Tumor Detection by Hybrid Quantum Classical Machine Learning via Thermal Imaging

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Abstract. Brain tumor detection is a critical task due to the brain's central role in regulating bodily functions and the potentially severe impact of undiagnosed tumors. Despite their effectiveness, traditional diagnostic methods like MRIs and CT scans are frequently costly, intrusive, and infrastructure-intensive. To enable quicker and easier tumor detection, we present a hybrid approach in this study that combines MobileNetV2, thermal imaging, and hybrid quantum classical machine learning. OpenCV is used to create a pipeline for simulating thermal imaging without the need for specialized hardware. To improve tumor localization, image enhancement techniques like edge detection and blurring are applied. The proposed model achieves an average accuracy of 98.2%, demonstrating competitive performance compared to classical deep learning models like CNN, VGG16, and DenseNet, while offering improved efficiency and applicability.

Keywords: Quantum computing, Deep learning, Brain tumor detection, Image processing, Thermal Imaging

1 Introduction and Literature Review

The brain is the most critical organ in the human body, governing thoughts, emotions, memory, and bodily functions through a network of neurons. It processes sensory information, regulates motor activity, and enables advanced functions like learning and reasoning. Any abnormalities, such as tumors, can significantly disrupt these vital processes.

A brain tumor is an abnormal mass that forms in the brain, potentially interfering with its normal functioning. Tumors can be benign or malignant, and their effective treatment depends heavily on timely and accurate diagnosis. Studies show that males are at a higher risk than females, with male-to-female risk ratios between 1.22 and 1.67 across tumor types. Additionally, racial disparities exist, with White individuals facing the highest risk for most tumor types. For example, the risk of glioblastoma is 4.07 in men and 2.55 in women [1].

Traditional diagnostic techniques like MRI, CT scans, and biopsies, although effective, are resource-intensive, costly, and often inaccessible in underdeveloped regions [2, 3]. This creates a pressing need for alternative solutions that are accurate, cost-effective, and accessible.

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Recent advances in Hybrid Quantum Classical Machine Learning (QML) present a promising solution. By combining quantum computing's speed and parallelism with classical image analysis, QML can handle high-dimensional image data efficiently, improving feature extraction, classification, and object detection [4, 5, 6, 7]. This has significant implications for medical imaging.

Several research efforts have explored this space. Quantum Machine Learning has gained traction for its potential to accelerate computation and improve model accuracy. Hybrid quantum-classical architectures, such as variational quantum circuits and quantum convolutional neural networks, have shown efficacy in complex tasks like medical diagnosis and image classification.

Rajesh et al.[4] developed Quantum Convolutional Neural Networks (QCNNs) for image recognition and object detection, achieving improved performance over classical models in domains like computer vision and cryptography.

Anupama et al.[5] proposed a hybrid quantum-classical Graph Neural Network (GNN) using a Variational Quantum Classifier (VQC) for cancer subtyping. Both fixed and end-to-end training variants demonstrated higher accuracy compared to classical GNNs, thanks to amplitude encoding and better training dynamics. Senokosov et al.[6] introduced two QML models for image classification. A hybrid model using parallel quantum circuits reached 99.21% accuracy on the MNIST dataset with fewer parameters than classical models, while a model with a Quantum layer achieved strong generalization with reduced complexity.

M Sadeghi et al.[8] proposed a new intra-tissue imaging (ITI) technique integrating artificial tactile sensing (ATS) and artificial neural networks (ANN) has been suggested for non-invasive detection and localization of brain tumors. The method's feasibility was tested by two simulations: (i) an in vitro situation with a heated agar tissue phantom to simulate heat generated due to tumor, and (ii) a parafalcine meningioma case from MRI-derived models to approximate tumor temperature and depth. A robot-assisted system touched the phantom to sense temperature changes, and ANN estimated the power and depth of the tumor. Results show that ATS-based ITI is a trustworthy method for tumor localization and diagnosis.

Liu et al.[7] designed a hybrid quantum-classical CNN (QCCNN) compatible with current quantum hardware. Their model preserved core CNN properties and outperformed its classical counterpart on the Tetris dataset.

Kumar et al.[9] employed a Quantum Support Vector Machine (QSVM) for classifying brain tumors using the Brats 2015 dataset. The QSVM achieved a 95% accuracy—1.6% higher than traditional SVMs—and demonstrated faster computation on both simulated and real quantum processors.

Novelty of the Work:

- Integration of the MobileNETV2 model, Hybrid Quantum Classical Machine Learning, and Thermal Imaging Techniques for brain tumor detection.
- Use of OpenCV to simulate thermal imaging effects without requiring actual thermal cameras.

- Demonstration of performance comparable or superior to conventional deep learning models such as CNN, VGG16, and DenseNet.

2 Background

Brain tumor detection has become more accurate and efficient through deep learning innovations, including CNN models, quantum machine learning, and imaging techniques.

2.1 Hybrid Quantum Classical Machine Learning

A hybrid quantum-classical model integrates classical deep learning architectures with quantum computing components to leverage the strengths of both paradigms. In this setup, the classical part—often a convolutional or dense neural network—performs feature extraction and initial data processing. The intermediate output is then passed to a quantum circuit, which uses qubits and quantum gates to apply transformations that capture complex correlations through quantum phenomena like superposition and entanglement.

The quantum circuit returns measurement-based outputs, which are combined with the classical network for further processing and final classification. This fusion enhances computational efficiency and allows the model to explore a richer solution space with fewer parameters. Such models are particularly useful in high-dimensional tasks like medical image classification, where subtle patterns, such as tumor boundaries, need precise identification.

2.2 MobileNetV2

MobileNetV2 is a lightweight deep learning architecture optimized for mobile and edge devices. Improves MobileNetV1 by using depthwise separable convolutions and introducing inverted residuals with linear bottlenecks.

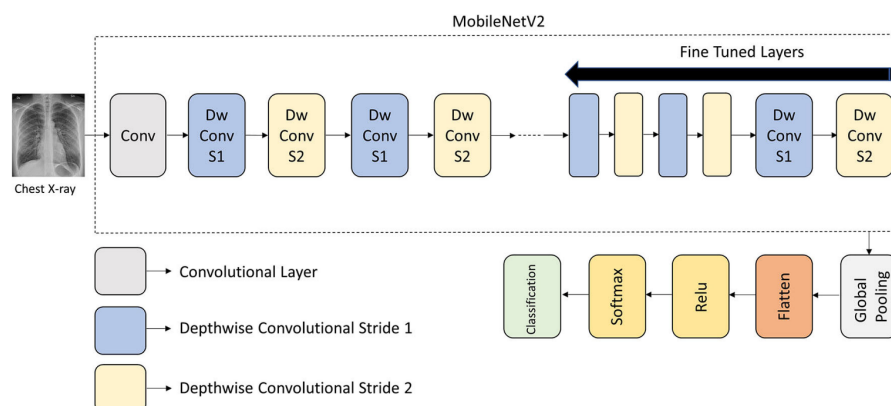


Figure: Functioning of MobileNETV2 detection model [10]

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Depthwise Separable Convolution: Standard convolution requires $D_K^2 \cdot M \cdot N \cdot D_F^2$ operations, whereas separable depth convolution reduces it to:

$$D_K^2 \cdot M \cdot D_F^2 + M \cdot N \cdot D_F^2$$

where D_K is the kernel size, D_F is the size of the feature map, and M, N are input/output channels.

Inverted Residual Block: Each block consists of: 1. 1x1 Expansion: Expands input by factor t . 2. 3x3 Depthwise Conv: Preserves spatial features. 3. 1x1 Linear Projection: Compresses back to lower dimensions.

Mathematically:

$$Y = W_3 \cdot \sigma(W_2 \cdot (W_1 \cdot X))$$

where W_1, W_2, W_3 are weights, σ is ReLU6, and X, Y are input/output tensors. For this work, we have taken ReLU.

2.3 Thermal Imaging

Thermal imaging captures temperature variations arising from the metabolic activity of tumors. These differences are recorded by infrared cameras and aid in early diagnosis. For improved visualization, OpenCV's `applyColorMap` with `COLORMAP_JET` was used to transform grayscale images into pseudocolor. This will help to visualize small changes within an image which sometime may not be possible in grayscale images.

2.4 Gaussian Blurring

Gaussian blurring is an operation that can be employed to minimize image detail and noise by smoothing the image. It is done by convolving the image with a Gaussian kernel that gives more weights to the pixels in the middle and progressively smaller weights to pixels farther away from the center. The Gaussian function employed for the operation is as follows:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right)$$

where σ controls the spread or intensity of the blur, and (x, y) are the coordinates representing the distance of each pixel from the center of the kernel.

2.5 Canny Boundary Detection:

Boundary detection is a crucial technique in image processing that involves identifying the outlines of objects by detecting significant changes in image intensity. The **Canny edge detection algorithm** is a widely-used and effective method for this purpose, consisting of the following steps:

Hybrid QCML for Brain Tumor Detection 5

1. **Gaussian Smoothing:** Gaussian filter $G(x, y, \sigma)$ is applied for smoothing and noise reduction. This is essential for enhancing edge detection accuracy:

$$G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right)$$

The smoothed image is obtained by convolution:

$$I_s(x, y) = I(x, y) * G(x, y, \sigma)$$

2. **Gradient Calculation:** This computing the image gradients $G_x(x, y)$ and $G_y(x, y)$ to identify areas with sharp intensity transitions. These gradients are then used to calculate the overall edge strength (magnitude) and orientation (direction):

$$G(x, y) = \sqrt{G_x(x, y)^2 + G_y(x, y)^2}, \quad \theta(x, y) = \arctan\left(\frac{G_y(x, y)}{G_x(x, y)}\right)$$

3. **Non-Maximum Suppression:** Preserves the most prominent edges, this suppresses all gradient values that are not considered local maxima along the direction of the gradient. This helps to thin out the edges, making them more precise.
4. **Double Thresholding:** The resulting edges are then classified based on two threshold values T_{high} and T_{low} . Strong edges clearly stand out, while weak edges are retained only if they fall within the acceptable range:

$$\text{Strong edges: } G(x, y) \geq T_{\text{high}}, \quad \text{Weak edges: } T_{\text{low}} \leq G(x, y) < T_{\text{high}}$$

5. **Edge Tracking by Hysteresis:** Finally, weak edges are either preserved or discarded based on their connectivity to strong edges. This ensures continuous and connected edge structures in the final edge map, reducing noise and false detections.

2.6 Quantum Gates

Quantum Gates are unitary operations that change the state of qubits, forming the basic components of quantum circuits. Some of the quantum gates are listed below-

- **RY Gate (Rotation around Y-axis)**

The RY gate performs a rotation of a qubit around the Y-axis by an angle θ .

$$RY(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

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– **RX Gate (Rotation around X-axis)**

The RX gate rotates a qubit around the X-axis by an angle θ .

$$RX(\theta) = \begin{bmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

– **RZ Gate (Rotation around Z-axis)**

The RZ gate rotates a qubit around the Z-axis by an angle θ .

$$RZ(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$$

– **CNOT Gate (Controlled-NOT)**

The CNOT gate flips the target qubit if the control qubit is in state $|1\rangle$.

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

These gates will help us to build a quantum circuit which we will use to make the Hybrid Quantum Classical Machine Learning model. We have used PennyLane to design the quantum circuit.

2.7 Hybrid Quantum-Classical Machine Learning vs Classical Deep Learning Models

Hybrid quantum-classical machine learning integrates the best of both quantum computing and classical machine learning. In this method, quantum algorithms are utilized to address intricate computations, allowing for a possible exponential speedup in some operations like optimization, encoding data, and feature extraction. Classical machine learning models, including deep neural networks, continue to perform the training and decision-making tasks, where classical hardware is best suited to manage enormous data sets and offer resilience.

Conversely, traditional deep learning models depend purely on traditional computing devices to carry out activities such as image classification, text processing, and speech recognition. While traditional deep learning models have become extremely successful across many fields, they tend to be constrained by available computational resources, particularly when handling large data sets or computationally intensive problems.

The hybrid method seeks to harness the special strengths of quantum computing, including superposition and entanglement, to possibly solve problems more quickly and efficiently than current deep learning systems. Yet it remains nascent, with issues regarding availability of hardware, error levels, and uniting quantum and classical systems. Classical deep learning is more developed and broadly supported, with successful experience in numerous practical uses.

In general, though classical deep learning models control today's machine learning tasks, hybrid quantum-classical models look to be bright stars for tomorrow,

with potential for a new computational efficiency paradigm in solving very complex problems.

3 Proposed Model

The workflow is depicted in Figure 3. The dataset consists of MRI images sourced from a Kaggle dataset, which are loaded into Google Colab. Libraries such as numpy, matplotlib, pandas, seaborn, os, pennylane, scikit-learn, and tensorflow are used for processing.

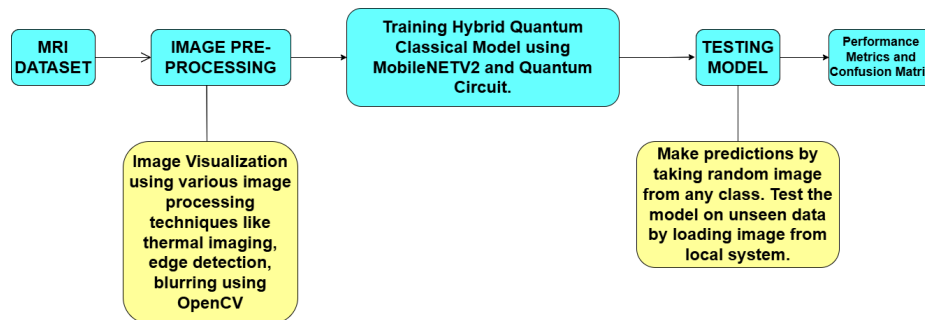


Figure: Proposed Workflow

Data visualization is performed using Thermal imaging, Gaussian blur, and Canny edge detection to determine suitable parameters for later processing. Images are labeled as 0 for healthy and 1 for tumor-affected cases. They are read in grayscale, resized to 128x128 pixels, normalized between 0 and 1, and split into 80% training and 20% testing data. The images are converted to RGB format to match the input requirements of MobileNetV2.

The model follows a hybrid quantum-classical approach. MobileNetV2, pre-trained on ImageNet with frozen weights, extracts feature representations. A global average pooling layer reduces dimensionality, followed by a dense layer with a specified number of neurons using ReLU activation.

The extracted features are passed to a quantum circuit, where RX, RY, and RZ rotations, along with CNOT entanglement, encode quantum correlations. Pauli-Z measurements are taken, and the real and imaginary components are processed through a quantum kernel that applies matrix multiplications with trainable parameters and a tanh activation function.

The quantum-processed data is fed into a classical neural network with 128 and 64 neurons using ReLU activation, defined as $\text{ReLU}(x) = \max(0, x)$, which introduces non-linearity by outputting zero for negative inputs. This is followed by a final sigmoid neuron for binary classification, where the sigmoid function is $\sigma(x) = \frac{1}{1+e^{-x}}$, mapping the output between 0 and 1. The model is compiled with the Adam optimizer and binary cross-entropy loss and is trained for 30 epochs. Then it is evaluated on the test dataset to determine accuracy. The model diagram is given below-

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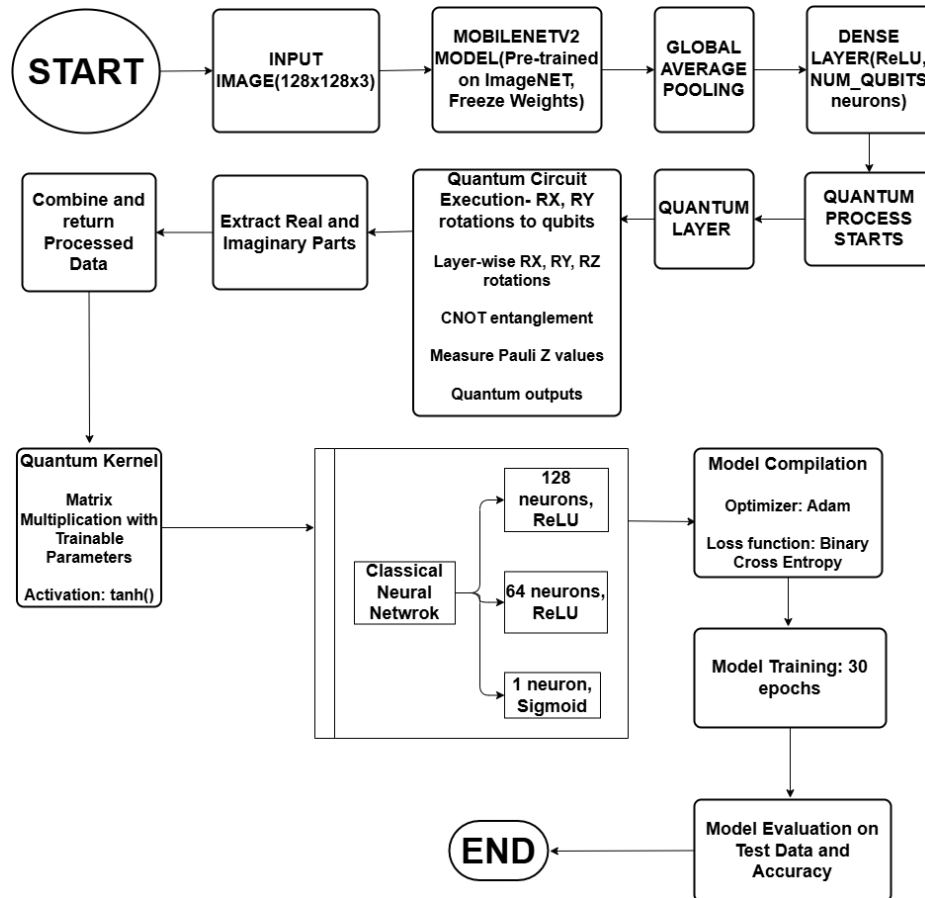


Figure: Diagram of the proposed system

4 Result

4.1 Dataset

For our work we have made our dataset, sourced from an existing Kaggle dataset[11]. The composition of the data set is as follows-

Category	Number of Images
No Tumor	1500
Tumor	
Glioma	500
Pituitary	500
Meningioma	500
Total	3000

4.2 Parameters and System Configuration

The MobileNetV2 model is trained with a batch size of 32 for 30 epochs using the Adam optimizer with a learning rate of 0.001. Binary cross-entropy is used as the loss function. For image preprocessing, a Gaussian blur with a 5x5 kernel and standard deviation of 1 is used. Canny edge detection is performed with upper and lower thresholds set to 20. COLORMAP_JET is applied from OpenCV for thermal visualization. The model's average validation accuracy is calculated over five compilations. The system is developed on Google Colab with computer of Intel i5 processor. T4 GPU is used for faster computation.

4.3 Results and discussion

Accuracy, precision, f1 score and recall are most important metrics that gives insights about how well a model is performing. Confusion matrix is used to visualize how many images are correctly and wrongly classified by any model. Below is the confusion matrix along with the necessary metrics for a single run of our proposed model.

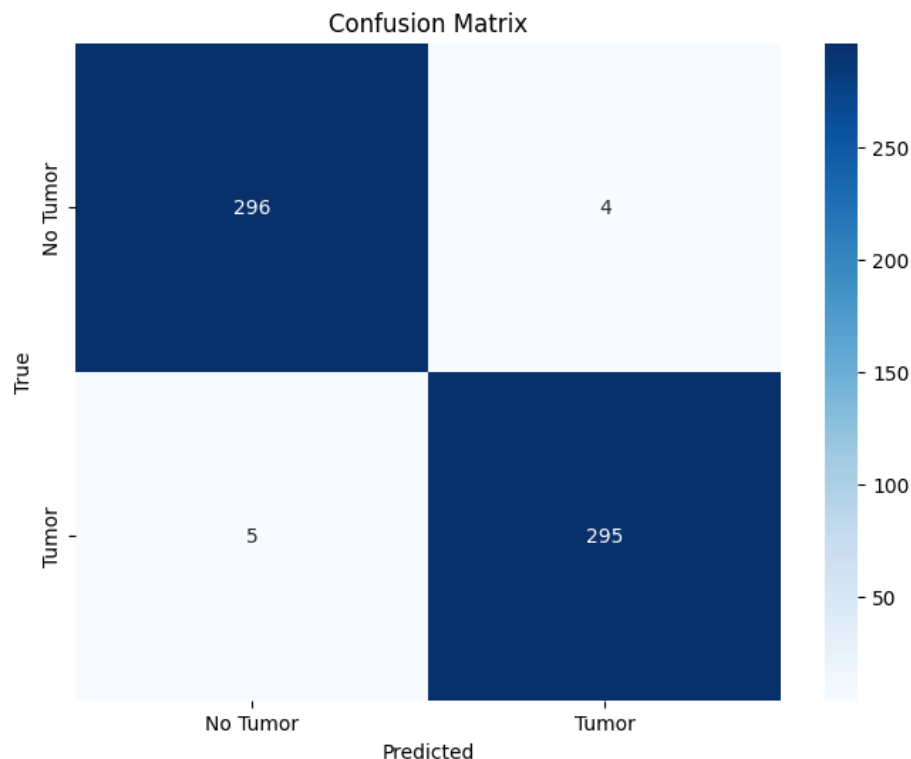


Figure: Confusion Matrix

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From the above confusion matrix we calculate the other metrics as well-

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{295}{295+4} = \frac{295}{299} \approx 0.987$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{295}{295+5} = \frac{295}{300} \approx 0.983$$

$$F1\text{-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \times \frac{0.987 \times 0.983}{0.987 + 0.983} \approx 0.985$$

$$\text{Accuracy} = \frac{TP+TN}{\text{Total_Samples}} = \frac{295+296}{295+296+4+5} = \frac{591}{600} \approx 0.985$$

From the above results, our model achieves an accuracy of 98.5% using Quantum Machine Learning. Below are the graphs for validation accuracy and validation loss-

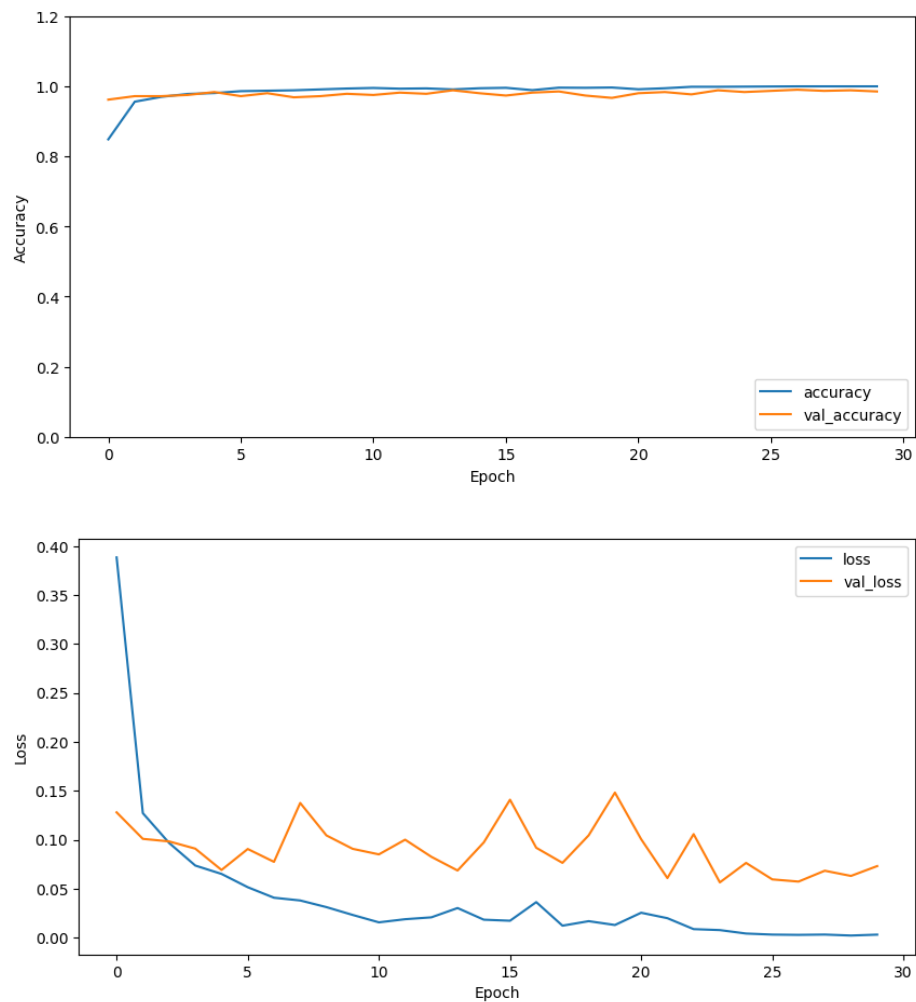


Figure:Accuracy and loss graphs

The model is tested on unseen data by uploading random brain MRI images to check model performance. The results are shown below in Fig 1:

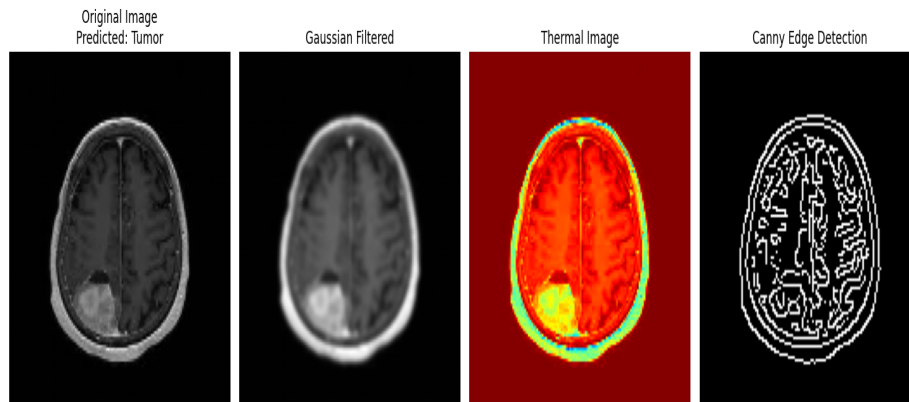


Fig. 1. Output after Image Processing

These image processing techniques ensure that even if a false prediction is made, one may still accurately determine whether the given image has tumor or not

5 Conclusion

This research focuses on developing a highly accurate model for detecting brain tumors and identifying affected regions in MRI scans using advanced image processing techniques. With an accuracy rate exceeding 98%, the model improves tumor detection, particularly in cases of false predictions. Future work could explore alternative CNN architectures, optimize hyperparameters, and increase the number of qubits in quantum computing models to enhance performance. Additionally, expanding the MRI image dataset would help the model generalize better to diverse cases, leading to even more precise and reliable predictions for clinical applications.

Disclosure of Interests. Authors declares no conflict of interest.

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